

Spectral Theory

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October 2024

Contents

1	Banach Spaces and bounded linear operators	2
2	Linear Operators on Hilbert Space	16

1 Banach Spaces and bounded linear operators

Definition 1.1. A norm on a vector space X is a function $\|\cdot\| : X \rightarrow [0, \infty)$ satisfying the following properties:

- (i) $\|x\| = 0$ if and only if $x = 0$ for all $x \in X$.
- (ii) $\|\lambda x\| = |\lambda| \|x\|$ for all $x \in X$ and $\lambda \in \mathbb{F}$.
- (iii) $\|x + y\| \leq \|x\| + \|y\|$ for all $x, y \in X$.

A vector space with a norm $\|\cdot\|$ is called a normed space denoted by $(X, \|\cdot\|)$.

Definition 1.2. Let $(X, \|\cdot\|)$ be a normed space, if it's complete in $\|\cdot\|$, we call X a Banach space. Completeness means for any Cauchy sequence in X , it also converges in X .

Example. We define the distance function $d : X \times X \rightarrow [0, \infty)$ can be defined as $d(x, y) := \|x - y\|$. This is a well-defined norm.

Definition 1.3. An inner product on a vector space X is a function $\langle \cdot, \cdot \rangle : X \times X \rightarrow \mathbb{F}$ satisfying the following properties:

- (i) $\langle x, y \rangle = \overline{\langle y, x \rangle}$ for all $x, y \in X$.
- (ii) $\langle \alpha_1 x + \alpha_2 y, z \rangle = \alpha_1 \langle x, z \rangle + \alpha_2 \langle y, z \rangle$ for all $\alpha_1, \alpha_2 \in \mathbb{F}$ and $x, y, z \in X$.
- (iii) $\langle x, x \rangle \geq 0$ and $\langle x, x \rangle = 0$ if and only if $x = 0$ for all $x \in X$.

Definition 1.4. A Hilbert space is a complete inner product with the norm $\|\cdot\| := \sqrt{\langle \cdot, \cdot \rangle}$.

Theorem 1.5. If $\|\cdot\|$ satisfies

$$\|f + g\| + \|f - g\| = 2(\|f\| + \|g\|) \text{ for all } f, g \in X$$

then $(X, \|\cdot\|)$ is also an inner space.

Examples.

- (1) Any finite normed space is a Banach space (i.e. $(\mathbb{R}^d, \|\cdot\|)$).
- (2) ℓ^p is a space of sequences $(x_k)_{k=1}^\infty$ satisfying

$$\|x\|_{\ell^p} = \left(\sum_{k=1}^{\infty} |x_k|^p \right)^{1/p} < \infty \text{ for } 1 \leq p < \infty$$

$$\|x\|_{\ell^\infty} = \sup |x_k| < \infty$$

ℓ^p is a Banach space for $1 \leq p \leq \infty$. To show the norm is well-defined, we need Minkowski inequality.

- (3) Let $X := C([a, b])$ with the norm $\|\cdot\|_{L^p}$ given by

$$\|f\|_{L^p} = \left(\int_a^b |f(t)|^p dt \right)^{1/p} \text{ for } 1 \leq p < \infty$$

$$\|f\|_{L^\infty} = \sup_{a \leq t \leq b} |f(t)|$$

then $(X, \|\cdot\|_{L^\infty})$ is a Banach space but $(X, \|\cdot\|_{L^p})$ for $1 \leq p < \infty$ is not a Banach space. Construct a sequence

$$f_n(t) = \begin{cases} 0 & 0 \leq t \leq \frac{1}{2} - \frac{1}{2n} \\ nt + (\frac{1}{2} - \frac{1}{2n}) & \frac{1}{2} - \frac{1}{2n} \leq t \leq \frac{1}{2} + \frac{1}{2n} \\ 1 & \frac{1}{2} + \frac{1}{2n} \leq t \leq 1 \end{cases}$$

then

$$\|f_n - f_m\|_{L^p}^p = \int_{\frac{1}{2} - \max\{\frac{1}{2n}, \frac{1}{2m}\}}^{\frac{1}{2} + \max\{\frac{1}{2n}, \frac{1}{2m}\}} |f_n(t) - f_m(t)|^p dt$$

Note that $0 \leq f_n \leq 1$ therefore $-1 \leq f_n - f_m \leq 1$ which means $|f_n - f_m| \leq 1$. In this case,

$$\|f_n - f_m\|_{L^p}^p = \max\left\{\frac{1}{n}, \frac{1}{m}\right\} \rightarrow 0 \text{ as } n, m \rightarrow \infty$$

therefore by definition of a Cauchy sequence, f_n is a Cauchy. If $(X, \|\cdot\|_{L^p})$ is a Banach space, then $f_n \rightarrow f \in X$. Suppose $f \in X$, for fixed $\delta > 0$ there exists $N > 0$ such that

$$\int_{\frac{1}{2} + \delta}^1 |f_n(t) - 1|^p dt = \int_{\frac{1}{2} + \delta}^1 |f(t) - f_n(t)|^p dt \leq \|f_n - f\|_{L^p}^p \rightarrow 0$$

Hence $f = 1$ on $[\frac{1}{2} + \delta, 1]$ for any $\delta > 0$ i.e. $\lim_{t \rightarrow \frac{1}{2}^+} f(t) = 1$. Similarly we can prove that $f = 0$ on $[\frac{1}{2} - \delta, 1]$ for any $\delta > 0$. Thus, we see that the limit is not continuous i.e. $(X, \|\cdot\|_{L^p})$ is not a Banach space.

(4) $X = C^k([a, b])$ with the norm

$$\|f\|_{C^k([a, b])} = \sum_{i=1}^k \sup_{a \leq t \leq b} |f^{(i)}(t)|$$

then $(X, \|\cdot\|_{C^k([a, b])})$ is a Banach space.

Definition 1.6. Let X, Y be normed spaces. A map $B : X \rightarrow Y$ is called a linear operator if B satisfies

$$B(\alpha_1 x_1 + \alpha_2 x_2) = \alpha_1 B(x_1) + \alpha_2 B(x_2) \text{ for all } \alpha_1, \alpha_2 \in \mathbb{F}, x_1, x_2 \in X$$

Theorem 1.7. Let X, Y be normed spaces, B be a linear map. Then the following statements are equivalent:

- (1) B is continuous.
- (2) B is continuous at 0.
- (3) There exists $c < \infty$, such that $\|Bx\| \leq c\|x\|$ for all $x \in X$.

Definition 1.8. If B is a bounded linear operator, we denote it as an element in $\mathcal{B}(X, Y)$ defined to be space of all bounded linear operator from X to Y . For $A \in \mathcal{B}(X, Y)$, we can define the following norm on $\mathcal{B}(X, Y)$ as

$$\|A\| := \inf\{c \in \mathbb{R} : \|Ax\| \leq c\|x\|, \forall x \in X\}$$

note that

$$\begin{aligned} \|A\| &\leq \inf\{c \in \mathbb{R} : \|Ax\| \leq c\|x\|, \forall \|x\| \leq 1\} \\ &\leq \inf\{c \in \mathbb{R} : \|Ax\| \leq c\|x\|, \forall \|x\| = 1\} \\ &\leq \sup\{\|Ax\| : \|x\| = 1, \forall x \in X\} \\ &= \sup\left\{\frac{\|Ax\|}{\|x\|} : \|x\| = 1, \forall x \in X\right\} \\ &\leq \sup\left\{\frac{\|Ax\|}{\|x\|} : \forall x \in X \setminus \{0\}\right\} \end{aligned}$$

But we also know that

$$\sup_{x \in X \setminus \{0\}} \frac{\|Ax\|}{\|x\|} \leq \|A\|$$

Thus we must have

$$\sup_{x \in X \setminus \{0\}} \frac{\|Ax\|}{\|x\|} = \|A\|$$

so the supremum is achieved on the unit ball.

Theorem 1.9. Let X, Y be normed spaces, $B \in \mathcal{B}(X, Y)$. If Y is complete then $(\mathcal{B}(X, Y), \|\cdot\|)$ is also a Banach space.

Definition 1.10. A vector space $\mathcal{E}$ is called an algebra if for all pairs $(x, y) \in \mathcal{E} \times \mathcal{E}$, there exists a unique element xy in $\mathcal{E}$ such that $x \cdot y = xy$. Also the algebra satisfies for all $x, y, z \in \mathcal{E}$ and $\lambda \in \mathbb{F}$:

- (i) $(x \cdot y) \cdot z = x \cdot (y \cdot z)$ (associative)
- (ii) $(x + y) \cdot z = x \cdot z + y \cdot z$ (right distributive)
- (iii) $x \cdot (y + z) = x \cdot y + x \cdot z$ (left distributive)
- (iv) $\lambda xy = (\lambda x) \cdot y = x \cdot (\lambda y)$

A normed algebra is an algebra satisfying

$$\|A \cdot B\| \leq \|A\| \|B\|$$

A Banach algebra is a complete normed algebra.

Remark. Let $\mathcal{B}(X) := \mathcal{B}(X, X)$. If X is a Banach space, then $\mathcal{B}(X)$ is also a Banach algebra satisfying

$$\|AB\| \leq \|A\| \|B\| \quad \forall A, B \in \mathcal{B}(X).$$

Definition 1.11. Let X be a Banach space. We say a series in X is convergent if its partial sum $s_j = \sum_{i=1}^j x_i$ is convergent i.e.

$$\lim_{j \rightarrow \infty} \|x - s_j\| = 0$$

then we write

$$\sum_{i=1}^{\infty} x_i = x$$

A series in X is called absolutely convergent if

$$\sum_{i=1}^{\infty} \|x_i\| < \infty$$

Lemma 1.12. An absolutely convergent series in X is convergent.

Definition 1.13. Let A be a bounded linear operator from X to Y . The kernel of A is defined as

$$\text{Ker}(A) := \{x \in X : Ax = 0\}$$

The range of A is defined as

$$\text{Ran}(A) := \{y \in Y : Ax = y \text{ for some } x \in X\}$$

In particular, $\text{Ker}(A) \subset X$ is a subspace of X and $\text{Ran}(A) \subset Y$ is a subspace of Y .

Remark. If X is Banach, Y is normed, then $\text{Ker}(A)$ is a closed subspace. However, this is not true to $\text{Ran}(A)$

Example. Let $X = C^1([a, b])$ and $Y = C([a, b])$, define $A := \frac{d}{dt}$

$$\|Af\| \leq C\|f\|$$

$$\sup_{a \leq t \leq b} \left| \frac{d}{dt} f(t) \right| \leq \sup_{a \leq t \leq b} \left| \frac{d}{dt} f(t) \right| + \sup_{a \leq t \leq b} |f(t)| = \|f\|_{C^1([a, b])}$$

Thus, A is bounded from $C^1([a, b])$ to $C([a, b])$.

Theorem 1.14 (Banach Theorem). Let X, Y be Banach spaces and $A \in \mathcal{B}(X, Y)$, if A is invertible then $A^{-1} \in \mathcal{B}(Y, X)$.

Proof. This follows immediately by the open mapping theorem. $\square$

Lemma 1.15. Let $A_l^{-1} : Y \rightarrow X$ and $A_r^{-1} : Y \rightarrow X$ defined by

$$A_l^{-1}A = I_X, \quad AA_r^{-1} = I_Y$$

then $A_l^{-1} = A_r^{-1}$.

Proof. Note that

$$A_l^{-1}AA_r^{-1} = A_r^{-1} \implies A_l^{-1} = A_r^{-1}$$

$\square$

Lemma 1.16.

(1) Let $A : X \rightarrow Y$ and $B : Y \rightarrow Z$ are invertible, then AB is invertible moreover

$$(AB)^{-1} = B^{-1}A^{-1}$$

(2) $A : X \rightarrow X$ is invertible and $B : X \rightarrow X$ commutes with A , then A^{-1} commutes with B .

(3) If $A : X \rightarrow X$ commutes with $B : X \rightarrow X$, then AB is invertible if and only if A and B are invertible.

Proof.

(2) If A is invertible, then there exists $D \in \mathcal{B}(X)$ such that

$$AD = DA = I$$

We want to show that

$$BD = DB$$

Note that $BD = DABD = DBAD = DB$ since A and B commutes and we are done.

(3) Suppose AB is invertible, then there exists $S \in \mathcal{B}(X)$ such that

$$SAB = I \implies (SA)B = I$$

i.e. B is invertible. In addition, since A and B commutes we also have A is invertible. Other direction is basically (1). $\square$

Lemma 1.17. Let X be a Banach space, $A \in \mathcal{B}(X)$. If $\|A\| < 1$, then $I - A$ is invertible and the inverse is given by

$$(I - A)^{-1} = \sum_{i=0}^{\infty} A^i$$

Moreover, $\|(I - A)^{-1}\| \leq \frac{1}{1 - \|A\|}$.

Proof. Let $s_j := \sum_{i=0}^j A^i$. Since $\sum_{i=0}^j \|A^i\| \leq \sum_{i=0}^k \|A\|^i < \infty$ since $\|A\| < 1$. Thus, s_j has a limit. Note that

$$(I - A)s_j = I - A^{j+1}$$

$$\implies \|(I - A)s_j - I\| = \| - A^{j+1} \| \leq \|A\|^{j+1} \rightarrow 0 \text{ as } j \rightarrow \infty$$

If $s_j \rightarrow s$ then $(I - A)s - I = 0$. By definition, $s = (I - A)^{-1}$. □

Lemma 1.18. Let $A, B \in \mathcal{B}(X)$ and A invertible. If $\|B\| < \frac{1}{\|A^{-1}\|}$ then $A + B$ is invertible and

$$\|(A + B)^{-1}\| \leq \frac{\|A^{-1}\|}{1 - \|A^{-1}\|\|B\|}$$

Proof. We can write $A + B = A(I + A^{-1}B)$ then by our previous Lemma we know that

$$(A + B)^{-1} = (I + A^{-1}B)^{-1}A^{-1} = \left(\sum_{i=0}^{\infty} (-A^{-1}B)^i\right)(A^{-1})$$

Now we see that

$$\begin{aligned} \|(A + B)^{-1}\| &\leq \frac{1}{1 - \|A^{-1}B\|} \|A^{-1}\| \\ &\leq \frac{\|A^{-1}\|}{1 - \|A^{-1}\|\|B\|} \end{aligned}$$
□

Definition 1.19. Let X, Y be Banach spaces. The resolvent set of a linear operator $B : X \rightarrow Y$ is the set of complex numbers λ such that $B - \lambda I$ is invertible and $(B - \lambda I)^{-1} \in \mathcal{B}(Y, X)$. The resolvent set of B is usually defined by $\rho(B)$. The spectrum of B is defined as $\sigma(B) := \mathbb{C} \setminus \rho(B)$.

Definition 1.20. A complex number λ is called an eigenvalue of B if there exists $x \in X \setminus \{0\}$ such that $Bx = \lambda x$ and x is called an eigenvector of B corresponding to λ .

Remark. Obviously, the set of eigenvalues of $B \subset \sigma(B)$. Let λ be an eigenvalue of B , then there exists $x \neq 0$ such that $Bx = \lambda x$. However, we have $B \cdot x = \lambda \cdot 0$. Therefore $B - \lambda I$ is not one to one and hence not invertible. Thus $\lambda \in \sigma(B)$.

Example. Consider $X = C([0, 1])$ with $\|\cdot\|_{L^\infty}$ norm. B is defined as

$$Bf(t) := t^2 f(t)$$

B is linear and bounded. If $\lambda \notin [0, 1]$ then $\frac{1}{t^2 - \lambda} \leq C$. Thus, $[0, 1]^c \subset \rho(B)$ i.e. $\sigma(B) \subset [0, 1]$. Let $\lambda \in [0, 1]$ then $B - \lambda I = t^2 - \lambda$, in particular $1 \in C([0, 1])$. Suppose $B - \lambda I$ is onto

$$(B - \lambda I)f = 1 \implies f = \frac{1}{t^2 - \lambda} \notin C([0, 1])$$

Thus $B - \lambda I$ is not onto which means $\sigma(B) = [0, 1]$. Suppose there is one $\lambda \in [0, 1]$ i.e. $Bf = \lambda f$ then $(t^2 - \lambda)f = 0$. Thus, $f = 0$ on $t^2 \neq \lambda$. Since $f \in C([0, 1])$ then $f = 0$ everywhere. Thus, f does not have an eigenvalue.

Theorem 1.21. Let $(X, \mathcal{A}, \mu)$ be a measure space, and let p satisfy $1 \leq p \leq +\infty$. Then $L^p(X, \mu)$ is complete under the norm $\|\cdot\|_{L^p}$.

Proof. First suppose that $p = +\infty$ and that $\{f_k\}_{k \in \mathbb{N}}$ is a sequence of functions that belong to $\mathcal{L}^\infty(X, \mu)$ and satisfy $\sum_{k \in \mathbb{N}} \|f_k\|_{L^\infty} < +\infty$. For each positive integer k let $N_k = \{x \in X \mid |f_k(x)| > \|f_k\|_{L^\infty}\}$. Then the series $\sum_{k \in \mathbb{N}} f_k(x)$ converges at each x outside $\bigcup_{k \in \mathbb{N}} N_k$ and the function f defined by

$$f(x) = \begin{cases} \sum_{k \in \mathbb{N}} f_k(x) & \text{if } x \notin \bigcup_{k \in \mathbb{N}} N_k, \\ 0 & \text{if } x \in \bigcup_{k \in \mathbb{N}} N_k \end{cases}$$

is bounded and $\mathcal{A}$ -measurable. Thus the inequality

$$\|f - \sum_{k=1}^n f_k\|_{L^\infty} \leq \sum_{k=n+1}^{\infty} \|f_k\|_{L^\infty}$$

holds for each n , and so

$$\lim_{n \rightarrow \infty} \|f - \sum_{k=1}^n f_k\|_{L^\infty} \leq \lim_{n \rightarrow \infty} \sum_{k=n+1}^{\infty} \|f_k\|_{L^\infty} = 0$$

Thus $L^\infty(X, \mu)$ is complete.

Now suppose that $1 \leq p < +\infty$ and that $\{f_k\}_{k \in \mathbb{N}}$ is a sequence of functions that belong to $\mathcal{L}^p(X, \mu)$ and satisfy

$\sum_{k \in \mathbb{N}} \|f_k\|_{L^p} < +\infty$. Define $g : X \rightarrow [0, +\infty]$ by

$$g(x) := \left(\sum_{k=1}^{\infty} |f_k(x)| \right)^p$$

Then by Minkowski's inequality

$$\left\| \sum_{k=1}^n |f_k| \right\|_{L^p} \leq \sum_{k=1}^n \|f_k\|_{L^p}$$

holds for each n , and so it follows from the monotone convergence theorem that

$$\int_X g d\mu = \lim_{n \rightarrow \infty} \int_X \left(\sum_{k=1}^n |f_k| \right)^p d\mu \leq \left(\sum_{k=1}^{\infty} \|f_k\|_{L^p} \right)^p$$

thus g is integrable. Consequently $g(x)$ is finite for almost every x and the series $\sum_{k \in \mathbb{N}} f_k(x)$ is absolutely convergent for almost every x . Define a function f on X by

$$f(x) = \begin{cases} \sum_{k=1}^{\infty} f_k(x) & \text{if } g(x) < +\infty, \\ 0 & \text{otherwise} \end{cases}$$

Then f is measurable and satisfies $|f|^p \leq g$, and so it belongs to $\mathcal{L}^p(X, \mu)$. Since

$$\lim_{n \rightarrow \infty} \left| \sum_{k=1}^n f_k(x) - f(x) \right| = 0$$

and

$$\left| \sum_{k=1}^m f_k(x) - f(x) \right|^p \leq g(x)$$

hold for almost every x , the dominated convergence theorem implies that

$$\lim_{n \rightarrow \infty} \left\| \sum_{k=1}^n f_k - f \right\|_{L^p} = 0$$

the completeness of $L^p(X, \mu)$ follows. □

Definition 1.22. Let X be Banach and $B_n \in \mathcal{B}(X)$ we say B_n converges to B strongly if for all $x \in X$, we have

$$B_n x \rightarrow Bx$$

i.e.

$$\|B_n x - Bx\|_X \rightarrow 0$$

We say B_n converges to B weakly if for all $g \in X^* := \mathcal{B}(X, \mathbb{F})$ and all $x \in X$

$$g(B_n x) \rightarrow g(Bx)$$

i.e.

$$|g(B_n x) - g(Bx)| \rightarrow 0$$

Definition 1.23. Let X be Banach and $B \in \mathcal{B}(X)$. A resolvent of B is an operator valued function then maps resolvent set $\rho(B)$ of B to $\mathcal{B}(X)$. Define via

$$\mathbb{C} \supset \rho(B) \ni \lambda \mapsto R(B, \lambda) = (B - \lambda I)^{-1} \in \mathcal{B}(X)$$

Lemma 1.24. Let X be a normed space and $z \in X$, there exists $g \in X^*$ such that $g(z) = \|z\|$ and $\|g\| = 1$.

Proof. Let $X_0 = \text{Span}(z) := \{\alpha z \in X \mid \alpha \in \mathbb{C}\}$. Then $X_0 \subset X$ and define g_0 as a linear functional on X_0 by

$$g_0(\alpha z) := \alpha \|z\|$$

$$|g_0(\alpha z)| = |\alpha| \|z\| \leq \|\alpha z\|$$

for all $x \in X_0$, we have $|g_0(x)| \leq \|x\|$. By Hahn-Banach Theorem, there exists $g \in X^*$ such that

$$|g(x)| \leq \|x\| \text{ for all } x \in X$$

and $g(x) = g_0(x)$ for all $x \in X_0$. Then we will have $\|g\| \leq 1$ but since $\frac{|g(z)|}{\|z\|} = 1$ so $\|g\| = 1$. □

Theorem 1.25. Let Z be a complex Banach space and Ω be an open subset of $\mathbb{C}$. Also let F be a Z -valued function that maps $\Omega \rightarrow Z$. Then the followings are equivalent:

(1) $\forall \lambda_0 \in \Omega$, $\frac{d}{d\lambda} F(\lambda)|_{\lambda=\lambda_0} = F'(\lambda_0) \in Z$ exists, i.e.

$$\lim_{\lambda \rightarrow \lambda_0} \left\| \frac{1}{\lambda - \lambda_0} (F(\lambda) - F(\lambda_0)) - F'(\lambda_0) \right\| = 0$$

(2) For all $\lambda_0 \in \Omega$, there exists an open neighbourhood of λ_0 such that $F(\lambda)$ can be expressed as

$$F(\lambda) = \sum_{i=0}^{\infty} F_n(\lambda_0)(\lambda - \lambda_0)^i, \quad F_n(\lambda_0) \in Z$$

(3) $\forall g \in Z^*$, the function

$$f(\lambda) := g(F(\lambda))$$

is holomorphic on Ω .

(4) If $z = \mathcal{B}(X, Y)$ for X, Y being Banach then (3) is equivalent to

$$\forall y^* \in Y^*, x \in X \text{ such that } \tilde{f}(\lambda) := y^*(F(\lambda)x) \in \mathbb{C}$$

is holomorphic in $\mathbb{C}$.

Lemma 1.26. For all $\lambda_1, \lambda_2 \in \rho(B)$ we have

$$R(B; \rho_0) - R(B; \rho_1) = (\rho_0 - \rho_1)R(B; \rho_0)R(B; \rho_1)$$

Proof. By definition

$$\begin{aligned} R(B; \lambda_0) - R(B; \lambda_1) &= R(B; \lambda_0)(I - (B - \lambda_0 I)R(B; \lambda_1)) \\ &= R(B; \lambda_0)((B - \lambda_1 I)R(B; \lambda_1) - (B - \lambda_0 I)R(B; \lambda_1)) \\ &= R(B; \lambda_0)(B - \lambda_1 I - (B - \lambda_0 I))R(B; \lambda_1) \\ &= (\lambda_0 - \lambda_1)R(B; \lambda_0)R(B; \lambda_1) \end{aligned}$$

Note that

$$\begin{aligned} \lim_{\lambda \rightarrow \lambda_0} \frac{1}{\lambda - \lambda_0} (R(B; \lambda) - R(B; \lambda_0)) &= \lim_{\lambda \rightarrow \lambda_0} R(B; \lambda)R(B; \lambda_0) = R^2(B; \lambda_0) \in \mathcal{B}(X) \\ \implies R(B, \lambda) \text{ is } \mathcal{B}(X) - \text{valued function on } \rho(B). \end{aligned}$$

□

Theorem 1.27. $-\lambda R(B; \lambda) \rightarrow I$ as $|\lambda| \rightarrow \infty$, and

$$\|R(B; \lambda)\| \geq \frac{1}{d(\lambda, \sigma(B))} \quad \forall \lambda \in \rho(B)$$

where $d(\lambda, \sigma(B)) := \inf_{\lambda_0 \in \sigma(B)} |\lambda - \lambda_0|$.

Proof. We want to show

$$\lim_{|\lambda| \rightarrow \infty} \|\lambda R(B; \lambda) - I\| = 0$$

for

$$\|B\| \leq \frac{1}{\|A^{-1}\|} = \frac{1}{\|(-\lambda z)^{-1}\|} = |\lambda|$$

we have

$$\|R(B; \lambda)\| \leq \frac{\|(-\lambda I)^{-1}\|}{1 - \|(-\lambda z)^{-1}\|\|B\|} = \frac{\frac{1}{|\lambda|}}{1 - \frac{1}{|\lambda|}\|B\|} = \frac{1}{|\lambda| - \|B\|}$$

Note that

$$\begin{aligned} \|\lambda R(B; \lambda) - I\| &= \|R(B; \lambda)(-\lambda) - R(B; \lambda)(B - \lambda I)\| = \|R(B; \lambda)(-\lambda I - B + \lambda I)\| \\ &\leq \|R(B; \lambda)\|\|B\| \leq \frac{\|B\|}{|\lambda| - \|B\|} \rightarrow 0 \end{aligned}$$

as $\lambda \rightarrow \infty$. $B - \lambda I = B - \lambda_0 I + (\lambda_0 - \lambda)I$. If $|\lambda - \lambda_0| = \|(\lambda - \lambda_0)I\| < \frac{1}{\|R(B; \lambda_0)\|} \implies B - \lambda I$ invertible then $\lambda \in \rho(B)$. Then

$$\lambda \notin \rho(B) \implies |\lambda - \lambda_0| \geq \frac{1}{\|R(B; \lambda_0)\|}$$

thus

$$\begin{aligned} |\lambda - \lambda_0| &\geq \frac{1}{\|R(B; \lambda_0)\|} \text{ for } \lambda_0 \in \rho(B), \lambda \in \sigma(B) \\ \|R(B; \lambda_0)\| &\geq \sup \frac{1}{|\lambda - \lambda_0|} = \frac{1}{\inf |\lambda - \lambda_0|} = \frac{1}{d(\lambda_0, \sigma(B))} \end{aligned}$$

□

Lemma 1.28. $\sigma(B)$ is compact for $B \in \mathcal{B}(X)$.

Proof. We first show the boundedness i.e. $\sigma(B) \subset \{\lambda : |\lambda| \leq \|B\|\}$. If $|\lambda| > \|B\|$ then by Lemma 1.18 we know that $B - \lambda I$ is invertible so $\lambda \in \rho(B)$ a contradiction. To show it's closed we would use the theorem above to show that $\rho(B)$ is open. □

Lemma 1.29. $\sigma(B) \neq \emptyset$ for all $B \in \mathcal{B}(X)$.

Proof. Suppose $\sigma(B) = \emptyset$, i.e. $\rho(B) = \mathbb{C}$. By Resolvent Theorem, for all $g \in X^*$, $x \in X$

$$f_{g,x}(\lambda) = g(R(B; \lambda)x)$$

is holomorphic on $\rho(B) = \mathbb{C}$. Moreover, we know that $f_{g,x} \rightarrow -\frac{g(x)}{\lambda} \rightarrow 0$. By Liouville's Theorem, $f_{g,x}(\lambda) = 0$ since it's bounded. From Hahn-Banach Lemma, $\exists \tilde{g} \in X$ such that

$$\tilde{g}(R(B; \lambda)x) = \|R(B; \lambda)x\|$$

then

$$\begin{aligned} f_{\tilde{g},x} &= \|R(B; \lambda)x\| = 0 \text{ for all } x \in X \\ \implies R(B; \lambda)x &= 0 \text{ for all } x \in X \end{aligned}$$

But by definition of resolvent, $(B - \lambda I)R(B; \lambda)x = Ix$. Now pick $x \neq 0$, the equation does not hold and we are done. $\square$

Definition 1.30. A spectral radius of $B \in \mathcal{B}(X)$ is defined as

$$r(B) := \sup\{|\lambda| \mid \lambda \in \sigma(B)\}$$

]

Definition 1.31. We define

$$\begin{aligned} \liminf_{n \rightarrow \infty} a_n &:= \lim_{n \rightarrow \infty} \inf_{k \geq n} a_n \\ \limsup_{n \rightarrow \infty} a_n &:= \lim_{n \rightarrow \infty} \sup_{k \geq n} a_n \end{aligned}$$

Theorem 1.32 (Spectral Radius Theorem).

$$r(B) = \lim_{n \rightarrow \infty} \|B^n\|^{\frac{1}{n}}$$

Proof. First observe that $r(B) \leq \|B\|$. In addition, since

$$B^n - \lambda^n I = (B - \lambda I)(B^{n-1} + \lambda B^{n-2} + \dots + \lambda^{n-1} I)$$

Recall from Lemma 1.16, if $\lambda \in \sigma(B)$, i.e. $B - \lambda I$ is not invertible so $B^n - \lambda^n I$ is not invertible i.e. $\lambda^n \in \sigma(B^n)$. Thus

$$\begin{aligned} r(B) &= \sup\{|\lambda| : \lambda \in \sigma(B)\} = (\sup\{|\lambda|^n : \lambda \in \sigma(B)\})^{\frac{1}{n}} \\ &\leq (\sup\{|z| : z \in \sigma(B^n)\})^{\frac{1}{n}} \\ &= r(B^n)^{\frac{1}{n}} \leq \|B^n\|^{\frac{1}{n}} \end{aligned}$$

i.e. $r(B) \leq \liminf_{n \rightarrow \infty} \|B^n\|^{\frac{1}{n}}$ for all $n \in \mathbb{N}$. We are left to prove $r(B) \geq \limsup_{n \rightarrow \infty} \|B^n\|^{\frac{1}{n}}$. Recall for $|\lambda| > \|B\|$, we have $(B - \lambda I)^{-1} = \sum_{i=0}^{\infty} ((\lambda I)^{-1} B)^i (\lambda I)^{-1}$. By resolvent theorem, we know that

$$(B - \lambda I)^{-1} = \sum_{i=0}^{\infty} \lambda^{-i-1} B^i$$

holds for all $\lambda \in \rho(B)$. Let

$$f(\lambda) := g(R(B; \lambda)x) \text{ for } g \in X^*, x \in X$$

then

$$f(\lambda) = \sum_{i=0}^{\infty} \lambda^{-i-1} g(B^i x)$$

Let $\lambda = ae^{i\theta}$ and $a > r(B)$ i.e. $ae^{i\theta} \in \rho(B)$.

$$\begin{aligned} \oint_C \lambda^n f(\lambda) d\lambda &= \int_0^{2\pi} (ae^{i\theta})^n f(ae^{i\theta}) dae^{i\theta} \\ &= i \int_0^{2\pi} a^{n+1} e^{i(n+1)\theta} f(ae^{i\theta}) d\theta \\ &= ia^{n+1} \int_0^{2\pi} e^{i(n+1)\theta} \sum_{k=0}^{\infty} (ae^{i\theta})^{-k-1} g(B^k x) d\theta \\ &= ia^{n+1} \sum_{k=0}^{\infty} \int_0^{2\pi} g(B^k x) a^{-k-1} e^{i(n-k)\theta} d\theta \\ &= ia^{n+1} g(B^n x) a^{-n-1} 2\pi = 2\pi i g(B^n x) \end{aligned}$$

Then

$$\begin{aligned}
|g(B^n x)| &\leq \frac{1}{2\pi} \left| \int_C \lambda^n f(\lambda) d\lambda \right| \\
&\leq \frac{1}{2\pi} \int_0^{2\pi} a^{n+1} |f(ae^{i\theta})| d\theta \\
&\leq \max_{\theta \in [0, 2\pi]} |f(ae^{i\theta})| a^{n+1} \\
&\leq \max_{\theta \in [0, 2\pi]} \|g\| \|R(B; ae^{i\theta})\| \|x\| a^{n+1}
\end{aligned}$$

By Hahn-Banach Lemma, there exists some $\tilde{g} \in X^*$ such that

$$\|B^n x\| = |\tilde{g}(B^n x)| \leq \max_{\theta \in [0, 2\pi]} \|\tilde{g}\| \|R(B; ae^{i\theta})\| \|x\| a^{n+1}$$

$$\|B^n x\| \leq \max_{\theta \in [0, 2\pi]} \|R(B; ae^{i\theta})\| \|x\| a^{n+1}$$

Thus, $\|B^n x\| \leq M a^{n+1}$ for some M and for all $n \in \mathbb{N}$. Then

$$\|B^n\|^{\frac{1}{n}} \leq a(a^{\frac{1}{n}} M^{\frac{1}{n}})$$

Taking lim sup yields

$$\limsup_{n \rightarrow \infty} \|B^n\|^{\frac{1}{n}} \leq a$$

This is true for all $a > r(B)$ so

$$\limsup_{n \rightarrow \infty} \|B^n\|^{\frac{1}{n}} \leq r(B)$$

and we are done. $\square$

Definition 1.33. Let $P(\eta) = \sum_{i=0}^N a_i \eta^i$ be a polynomial with $a_N \neq 0$. We define for $B \in \mathcal{B}(X)$

$$P(B) := \sum_{i=0}^N a_i B^i \in \mathcal{B}(X)$$

Theorem 1.34 (Spectral mapping theorem).

$$\sigma(P(B)) = P(\sigma(B)) := \{P(\lambda) : \lambda \in \sigma(B)\}$$

Proof. Observe $P(\eta) - P(\lambda) = (\eta - \lambda)Q(\eta; \lambda)$ where Q is also a polynomial. In particular, $P(B) - P(\lambda)I = (B - \lambda I)Q(B; \lambda)$. If $\lambda \in \sigma(B)$, then $B - \lambda I$ is not invertible which means $P(B) - P(\lambda)I$ is not invertible. Therefore, $P(\lambda) \in \sigma(P(B))$ i.e. $P(\sigma(B)) = \{P(\lambda) : \lambda \in \sigma(B)\} \subset \sigma(P(B))$.

Let $\mu \in \sigma(P(B))$, observe

$$P(B) - \mu = a_N (B - \tau_1(\mu)) \cdots (B - \tau_k(\mu))$$

If $\prod_i A_i$ is not invertible then there exists some A_I which is not invertible. Say $(B - \tau_I(\mu))$ is not invertible for $I \in \{1, \dots, k\}$. Thus there exists some $\tau_I(\mu) \in \sigma(B)$. In this case, $\mu \in \{P(\tau_I) : \tau_I \in \sigma(B)\} \subset P(\sigma(B))$. $\square$

Definition 1.35. Let $A \in \mathcal{B}(X, Y)$, the adjoint of the operator A is given by the following: $A^* : Y^* \rightarrow X^*$ such that for $g \in Y^*$ and $x \in X$

$$\mathbb{F} \ni A^*(g)(x) := g(Ax)$$

In particular, $A^* \in \mathcal{B}(Y^*, X^*)$.

Remark. Let $\tilde{x} \in X^* := \mathcal{B}(X, \mathbb{F})$. The norm is defined by

$$\|\tilde{x}\|_{X^*} := \sup\{|\tilde{x}(t)| : \|t\| = 1, t \in X\}$$

then

$$\begin{aligned}
\|A^*\| &= \sup\{\|A^*(g)\|_{X^*} : \|g\|_{Y^*} = 1, g \in Y^*\} \\
&= \sup\{\sup\{|A^*(g)(t)| : \|t\|_X = 1, t \in X\} : \|g\|_{Y^*} = 1, g \in Y^*\} \\
&= \sup\{\sup\{|g(A(t))| : \|t\|_X = 1, t \in X\} : \|g\|_{Y^*} = 1, g \in Y^*\} \\
&\leq \sup\{\sup\{\|g\|_{Y^*} \|A\| \|t\|_X : \|t\|_X = 1, t \in X\} : \|g\|_{Y^*} = 1\} \\
&= \sup\{\|g\|_{Y^*} \|A\| : \|g\|_{Y^*} = 1, g \in Y^*\} = \|A\|
\end{aligned}$$

In addition,

$$\|A\| = \sup\{\|At\|_Y : \|t\|_X = 1, t \in X\}$$

Recall from Hahn-Banach Lemma, there exists $g \in Y^*$ such that

$$g(At) = \|At\|_Y$$

Thus,

$$\begin{aligned}
\|A\| &= \sup\{\|At\|_Y : \|t\|_X = 1, t \in X\} \\
&\leq \sup\{|g(At)| : \|t\|_X = 1, t \in X, \|g\|_{Y^*} = 1, g \in Y^*\} \\
&= \sup\{|A^*(g)(t)| : \|t\|_X = 1, t \in X, \|g\|_{Y^*} = 1, g \in Y^*\} \\
&\leq \sup\{\|A^*\| \|g\|_{Y^*} \|t\|_X, \|t\|_X = 1, t \in X, \|g\|_{Y^*} = 1, g \in Y^*\} = \|A^*\|
\end{aligned}$$

Thus, $\|A\| = \|A^*\|$.

Example. Let $X = \mathbb{C}^2$ and $Y = \mathbb{C}^3$ and

$$M = \begin{pmatrix} 1 & 2i \\ 3i & 4 \\ 0 & 0 \end{pmatrix}$$

Its adjoint is given by

$$M^* = \begin{pmatrix} 1 & -3i & 0 \\ -2i & 4 & 0 \end{pmatrix}$$

Then $\ker(M^*) = \{(0, 0, y)^T : y \in \mathbb{C}\}$. Thus, $Au = f$ has a solution then $g(f) = 0$ for all $g \in \ker(M^*)$. The $\text{Ran}(A) \subset \{g \in Y : g(f) = 0 \ \forall g \in \ker(A^*)\}$.

Lemma 1.36. $\overline{\text{Ran}(A)} \subset \{g \in Y : g(f) = 0 \ \forall g \in \ker(A^*)\}$.

Proof. Let $y_n = A_x x$ be a Cauchy sequence in $\overline{\text{Ran}(A)}$. For any $g \in \ker(A^*)$, we have

$$\begin{aligned}
g(y) &:= g\left(\lim_{n \rightarrow \infty} y_n\right) = g\left(\lim_{n \rightarrow \infty} A x_n\right) \\
&= \lim_{n \rightarrow \infty} g(A x_n) = \lim_{n \rightarrow \infty} (A^* g)(x_n) \\
&= 0
\end{aligned}$$

i.e. $y \in \{f \in Y : g(f) = 0 \ \forall g \in \ker(A^*)\}$. □

Corollary 1.37. If $Au = f$ has a solution, then $g(f) = 0$ for all $g \in \ker(A^*)$.

Lemma 1.38. Let X_0 be a closed subspace of X and $X_0 \neq X$. Let $x_1 \in X \setminus X_0$, there exists $g \in X^*$ such that $g(x_1) = 1$ and $g(x_0) = 0$ for all $x_0 \in X_0$. Moreover,

$$\|g\| = \frac{1}{d}, \quad d = \text{dist}(x_1, X_0) = \inf_{x_0 \in X_0} \|x_1 - x_0\| > 0$$

Proof. Let $X_1 = \text{Span}\{x_1\} \oplus X_0 \subset X$. Define $\tilde{g}$ on X_1 such that $\tilde{g}(\alpha x_1 + x_0) := \alpha$. By definition,

$$\begin{aligned}
\|\tilde{g}\|_{X_1} &:= \sup_{\alpha x_1 + x_0 \in X_1} \frac{|\tilde{g}(\alpha x_1 + x_0)|}{\|\alpha x_1 + x_0\|} = \sup_{\substack{\alpha x_1 + x_0 \in X_1 \\ \alpha \neq 0}} \frac{|\alpha|}{\|\alpha x_1 + x_0\|} \\
&= \sup_{\substack{\alpha x_1 + x_0 \in X_1 \\ \alpha \neq 0}} \frac{1}{\|x_1 + \frac{x_0}{\alpha}\|} = \frac{1}{\inf_{\omega \in X_0} \|x_1 - \omega\|} = \frac{1}{d} > 0
\end{aligned}$$

By Hahn-Banach, there exists g extends $\tilde{g}$ i.e.

$$g(x) = \tilde{g}(x) \ \forall g \in X_1 \text{ and } \|g\| = \|\tilde{g}\| = 1 = \frac{1}{d}$$

□

Lemma 1.39. $\overline{\text{Ran}(A)} = \{g \in Y : g(f) = 0 \ \forall g \in \ker(A^*)\} =: \tilde{Y}$.

Proof. Suppose $\overline{\text{Ran}(A)} \neq \tilde{Y}$, there exists $y_1 \in \tilde{Y} \setminus \overline{\text{Ran}(A)} \subset Y \setminus \overline{\text{Ran}(A)}$. By previous Lemma, there exists $g \in Y^*$ such that $g(y_1) = 1$ and $g(y_0) = 0$ where $y_0 \in \overline{\text{Ran}(A)}$. This implies $0 = g(Ax) = A^*(g)(x)$ for all x i.e. $A^*(g) = 0 \implies g \in \ker(A^*)$ so $g(y_1) = 0$. Thus, we have shown the converse inclusion. □

Definition 1.40. An operator P in $\mathcal{B}(X)$ is called a projection if it is idempotent i.e. $P^2 = P$. Let $Q = I - P$, in particular, $Q^2 = Q$ so it's a projection as well.

Lemma 1.41. $\text{Ran}(P) = \ker(Q)$, $\text{Ran}(Q) = \ker(P)$. Moreover, $X = \ker(P) \oplus \text{Ran}(P)$.

Proof. Let $x \in \text{Ran}(P)$ i.e. $x = Px'$ for some $x' \in X$. Then

$$\begin{aligned}
Qx &= QPx' = (I - P)Px' = (P - P^2)x' = 0 \\
&\implies x \in \ker(Q)
\end{aligned}$$

i.e. $\text{Ran}(P) \subset \ker(Q)$. Let $y \in \ker(Q)$ i.e. $Qy = 0$ then

$$(I - P)y = y - Py = 0$$

so $y \in \text{Ran}(P)$ thus $\ker(Q) \subset \text{Ran}(P)$. Similarly, we can show that $\text{Ran}(Q) = \ker(P)$.

To prove $X = \ker(P) \oplus \text{Ran}(P)$. Suppose $x \in \ker(P) \cap \text{Ran}(P) \implies Px = 0$ and $Qx = 0$ i.e. $x = Px + Qx = 0$ so $\ker(P) \cap \text{Ran}(P) = \{0\}$.

In addition, since $x = Px + (I - P)x = Qx + Px \in \text{Ran}(Q) + \text{Ran}(P) = \ker(P) + \text{Ran}(P)$. $\square$

Remark. 0 and I are called trivial projections.

Theorem 1.42. If P is not a trivial projection, then

$$\sigma(P) = \{0, 1\}$$

Proof. By the Spectral Mapping theorem,

$$\sigma(P^2 - P) = \sigma(0) = \{0\}$$

i.e. $\lambda^2 - \lambda = 0 \implies \lambda = 0$ or 1 . Thus, $\sigma(P) \in \{0, 1\}$. Moreover, at least one of $0, 1 \in \sigma(P)$. We can show that $0, 1 \in \sigma(P)$. $\square$

Remark. For $A^2 + 2A = B$ if we know the spectrum of B then we know some information of A .

Definition 1.43. A subset $K \subset X$ is called relatively compact if every sequence in K has a Cauchy subsequence. And K is called compact if every sequence in K has a Cauchy subsequence whose limit is in K .

Proposition 1.44.

- (1) K be relatively compact implies K being bounded.
- (2) K be compact implies K being closed and bounded.
- (3) Every subset of relatively compact set is relatively compact.
- (4) Every closed subset of a compact set is compact.

Theorem 1.45. In a finite dimensional space X , all norms are equivalent i.e. let $\|\cdot\|_1$ and $\|\cdot\|_2$ be any two norms in X then there exists c_1 and c_2 such that

$$c_1\|x\|_1 \leq \|x\|_2 \leq c_2\|x\|_1 \quad \forall x \in X$$

Theorem 1.46. Relatively compact sets are equivalent to totally bounded set i.e. for all $\varepsilon > 0$ there exists a finite set N_ε such that

$$K \subset B_\varepsilon(N_\varepsilon)$$

Corollary 1.47. Every finite dimensional sub-linear space of a normed space X is closed.

Proof. Let $e_1, \dots, e_m$ be a basis for $X_0 \subset X$ with norm $\|\cdot\|$ then let

$$\|x\|' := \max_i |c_i|, \quad x = \sum_{i=1}^m c_i e_i$$

Every Cauchy sequence in $\|\cdot\|$ with $\|x_n - x_m\| < \varepsilon$ then

$$\|x_n - x_m\|' < c_2 \varepsilon$$

x_n is also a Cauchy sequence in $\|\cdot\|'$ which implies $x = \lim_{n \rightarrow \infty} x_n$ exists in $\|\cdot\|'$ by closeness of real numbers. $\square$

Example. Let X be infinite dimensional and let

$$d(x, y) := \begin{cases} 1 & x \neq y \\ 0 & x = y \end{cases}$$

Consider $\varepsilon = \frac{1}{2}$ then N_ε is a collection of x which is not finite.

Definition 1.48. A linear operator T from normed space X to normed space Y is called compact if T maps any bounded set in X to relatively set in Y .

Definition 1.49. An operator T is called compact is T maps bounded subsets of X to a relative compact set in Y . In particular $T(S_X)$ is a bounded set where $S_X := \{x \in X : \|x\| = 1\}$. Then

$$\|T\| =: \sup\{\|Tx\| : \|x\| \leq 1\}$$

is bounded. We denote the set of compact operator from X to Y by

$$Com(X, Y) \subset \mathcal{B}(X, Y)$$

Proof. We show the equivalence of the definitions.

($\Rightarrow$) : This is trivial.

($\Leftarrow$) : We start with $M \subset X$ be a bounded set. Then there exists some $\alpha > 0$ such that $M \subset \alpha S_X$ i.e. $\frac{M}{\alpha} \subset S_X$. Then $T(\frac{M}{\alpha})$ is a relatively compact set so $T(M)$ is relatively compact. $\square$

Definition 1.50. A linear operator T from X to Y is called finite rank operator if $\text{Ran}(T)$ is finite dimensional.

Remark. If $T \in \mathcal{B}(X, Y)$ is a finite rank operator, then T is compact.

Example. Let $A : \ell^p \rightarrow \ell^p$ be $A(x_1, x_2, \dots) = (x_1, x_2, 0, \dots)$ which is a finite rank operator.

Theorem 1.51 (Properties of compact operators).

- (1) $T_1 \in Com(X, Y)$ and $T_2 \in Com(X, Y)$ then $\alpha T_1 + \beta T_2 \in Com(X, Y)$ for $\alpha, \beta \in \mathbb{F}$.
- (2) $T \in Com(X, Y)$, $A \in \mathcal{B}(Y, X)$, $B \in \mathcal{B}(Z, X)$ then
$$AT \in Com(X, Z) \text{ and } TB \in Com(Z, Y).$$
- (3) $(T_n)_{n \in \mathbb{N}} \in Com(X, Y)$ and if $\|T - T_n\| \rightarrow 0$, then T is also compact.

Proof.

- (1) Let $(x_n)_{n \in \mathbb{N}}$ be any bounded sequence then we want to show $\alpha T_1 x_{n_j} + \beta T_2 x_{n_j}$ is Cauchy. By compactness of T_1 , we know that $(T_1 x_n)_{n \in \mathbb{N}}$ has a Cauchy subsequence $(T_1 x_{n_j})_{j \in \mathbb{N}}$. Consider $(x_{n_j})_{j \in \mathbb{N}}$ which is bounded, by compactness of T_2 we know that $T_2 x_{n_j}$ has a Cauchy subsequence say $(T_2 x_{n_{j_k}})_{k \in \mathbb{N}}$. Then

$$\alpha T_1 x_{n_{j_k}} + \beta T_2 x_{n_{j_k}}$$

is also Cauchy and we are done.

- (2) Let $(x_n)_{n \in \mathbb{N}}$ be a bounded sequence. Since T is compact, we know that $(Tx_n)_{n \in \mathbb{N}}$ has a Cauchy sequence $(Tx_{n_j})_{j \in \mathbb{N}}$ if and only if $\|Tx_{n_j} - Tx_{n_{j'}}\| \rightarrow 0$. Note that

$$\|ATx_{n_j} - ATx_{n_{j'}}\| \leq \|A\| \|Tx_{n_j} - Tx_{n_{j'}}\| \rightarrow 0$$

i.e. $(ATx_{n_j})_{j \in \mathbb{N}}$ is Cauchy.

Let M be a bounded set of Z . Since B is bounded, $B(M)$ is also bounded set of X . But since T is compact, $T(B(M))$ is relatively compact.

- (3) Let $(x_n)_{n \in \mathbb{N}}$ be a bounded sequence. In particular $\|x_n\| \leq M$. Let $\varepsilon > 0$ then by $\|T - T_k\| \rightarrow 0$, we know there exists $K > 0$ such that

$$\|T - T_k\| \leq \frac{\varepsilon}{3M} \Leftrightarrow \|Tx_n - T_k x_n\| \leq \frac{\varepsilon}{3M} \|x_n\| \leq \frac{\varepsilon}{3} \text{ for all } k > K$$

By compactness of T_k , we know that $T_k x_n$ has a Cauchy subsequence $T_k x_{n_j}$ such that

$$\|T_k x_{n_j} - T_k x_{n_{j'}}\| \leq \frac{\varepsilon}{3}, \quad \text{for all } j, j' > J$$

Note that

$$\begin{aligned} \|Tx_{n_j} - Tx_{n_{j'}}\| &= \|Tx_{n_j} - T_k x_{n_j} + T_k x_{n_j} - T_k x_{n_{j'}} + T_k x_{n_{j'}} - Tx_{n_{j'}}\| \\ &\leq \|Tx_{n_j} - T_k x_{n_j}\| + \|T_k x_{n_j} - T_k x_{n_{j'}}\| + \|T_k x_{n_{j'}} - Tx_{n_{j'}}\| \\ &\leq \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon \end{aligned}$$

and we are done. $\square$

Lemma 1.52. Let X_0 be a closed subspace of a normed space X and $X_0 \neq X$. For $\varepsilon > 0$, then there exists a vector $x_1 \in X \setminus X_0$ such that

$$\|x_1 - x_0\| \geq 1 - \varepsilon \text{ for all } x_0 \in X_0.$$

and $\|x_1\| = 1$.

Proof. If $\varepsilon \geq 1$, this is trivial.

Let $0 < \varepsilon < 1$ then x_0 is closed implies

$$d := \inf_{x_0 \in X_0} \|\tilde{x}_1 - x_0\| > 0$$

There exists some $\tilde{x}_0 \in X_0$ such that $d \leq \|\tilde{x}_1 - \tilde{x}_0\| \leq \frac{d}{1-\varepsilon}$. Then let

$$x_1 := \frac{\tilde{x}_1 - \tilde{x}_0}{\|\tilde{x}_1 - \tilde{x}_0\|}$$

Then

$$\|x_1 - x_0\| = \left\| \frac{\tilde{x}_1 - \tilde{x}_0}{\|\tilde{x}_1 - \tilde{x}_0\|} - x_0 \right\| = \frac{1}{\|\tilde{x}_1 - \tilde{x}_0\|} \|\tilde{x}_1 - \|\tilde{x}_1 - \tilde{x}_0\|x_0\|$$

Note that $\|\tilde{x}_1 - \tilde{x}_0\|x_0 \in X_0$, thus

$$\frac{1}{\|\tilde{x}_1 - \tilde{x}_0\|} \|\tilde{x}_1 - \|\tilde{x}_1 - \tilde{x}_0\|x_0\| \geq \frac{1}{\|\tilde{x}_1 - \tilde{x}_0\|} d \geq 1 - \varepsilon$$

and we are done. $\square$

Theorem 1.53. Every bounded subset of X is relatively compact if and only if X is finite dimensional.

Proof. ($\Leftarrow$) : Known from Analysis 4.

($\Rightarrow$) : Proof by contrapositive statement. Let X be infinite dimensional. Let $x_0 \in X$ such that $\|x_0\| = 1$. Consider $X_0 = \text{Span}\{x_0\}$ so X_0 is closed i.e. $X \neq X_0$. By Lemma above, there exists $x_1 \in X \setminus X_0$ such that $\|x_1\| = 1$ and $\|x_1 - x_0\| \geq \frac{1}{2}$ for all $x_0 \in X_0$. Let $X_1 := \text{Span}(x_1) \oplus X_0$ is closed and not equal to X . And we will do this repeatedly such that

$$X_k = \text{Span}(x_k) \oplus X_{k-1}$$

and $\|x_k - x_{k-1}\| \geq \frac{1}{2}$ for all $x_{k-1} \in X_{k-1}$. We have constructed $(x_n)_{n \in \mathbb{N}}$ a bounded sequence but it does not have a Cauchy subsequence. Therefore, it implies X is not relatively compact. $\square$

Corollary 1.54. Identity operator I_X is compact if and only if X is finite dimensional.

Proof. Since $I_X(M) = M$, if I_X is compact then X is finitely dimensional. $\square$

Corollary 1.55. Let X and Y be Banach and $T \in \text{Com}(X, Y)$. If one of X or Y is infinitely dimensional, then T is not invertible i.e. $0 \in \sigma(T)$.

Proof. Suppose T is invertible, then $T^{-1} \in \mathcal{B}(Y, X)$ by Banach theorem. Then $I_X = T^{-1}T$ then I_X is compact as well. But I_X is compact if and only if X is finite dimensional. Similarly, we could prove that Y is finite dimensional. $\square$

Corollary 1.56. If X is infinitely dimensional Banach space and $T \in \text{Com}(X)$, then $0 \in \sigma(T)$.

Proof. Immediate. $\square$

Theorem 1.57. Let λ be a non-zero eigenvalue of T and $T \in \text{Com}(X)$. Denote by X_λ the eigenspace for λ . Then X_λ is finite dimensional.

Proof. Let M be a bounded subset of X_λ . Let $(x_n)_{n \in \mathbb{N}} \subset M$. Since $\lambda \neq 0$, we have

$$x_n = \frac{Tx_n}{\lambda}$$

But T is compact, $(Tx_n)_{n \in \mathbb{N}}$ has a Cauchy subsequence $(Tx_{n_j})_{j \in \mathbb{N}}$ i.e.

$$x_{n_j} = \frac{Tx_{n_j}}{\lambda}$$

is Cauchy i.e. M is relatively compact. Thus, X_λ is finite dimensional. $\square$

Theorem 1.58. Let X be a Banach space and $T \in \text{Com}(X)$. $\lambda \neq 0$ is not an eigenvalue of T , then there exists $c > 0$ such that

$$\|(T - \lambda)x\| \geq c\|x\| \text{ for all } x \in X$$

Proof. Suppose for all $c > 0$ there exists $x \in X$ such that $\|(T - \lambda)x\| < c\|x\|$. Let $c_n = \frac{1}{n}$ there exists $x_n \in X$ such that

$$\|(T - \lambda)x_n\| < \frac{1}{n}\|x_n\|$$

There exists $(x_n)_{n \in \mathbb{N}} \in X$ such that $\|x_n\| = 1$ and $\|(T - \lambda)x_n\| \leq \frac{1}{n}$. Since T is compact, $(Tx_n)_{n \in \mathbb{N}}$ has Cauchy subsequence i.e. $(Tx_{n_j})_{j \in \mathbb{N}}$ is Cauchy. Without loss of generality, assume

$$\|Tx_{n_j} - Tx_{n_k}\| \leq \frac{1}{\min\{n_j, n_k\}}$$

Note that since $\lambda \neq 0$

$$\begin{aligned} \|x_{n_j} - x_{n_k}\| &= \|x_{n_j} + \frac{T}{\lambda}x_{n_j} - \frac{T}{\lambda}x_{n_j} - x_{n_k} + \frac{T}{\lambda}x_{n_k} - \frac{T}{\lambda}x_{n_k}\| \\ &= \left\| \frac{1}{\lambda}(Tx_{n_j} - Tx_{n_k}) + \frac{1}{\lambda}(\lambda x_{n_j} - Tx_{n_j}) + \frac{1}{\lambda}(Tx_{n_k} - \lambda x_{n_k}) \right\| \\ &\leq \frac{1}{|\lambda|} \left(\frac{1}{\min\{n_j, n_k\}} + \frac{1}{n_j} + \frac{1}{n_k} \right) \rightarrow 0 \end{aligned}$$

as $n_j, n_k \rightarrow 0$. Thus, x_{n_j} is also Cauchy. Since X is Banach, $x_{n_j} \rightarrow \tilde{x} \in X$. In particular,

$$\begin{aligned}\lambda \tilde{x} &= \lim_{j \rightarrow \infty} \lambda x_{n_j} = \lim_{j \rightarrow \infty} T x_{n_j} = T \lim_{n \rightarrow \infty} x_{n_j} = T \tilde{x} \\ &\implies \|\tilde{x}\| = 1\end{aligned}$$

i.e. λ is an eigenvalue of T . □

Lemma 1.59. Let X be Banach and $B \in \mathcal{B}(X, Y)$ and there exists $c > 0$ such that $\|Bx\| \geq c\|x\|$ for all $x \in X$. Then $\ker B = \{0\}$ and $\text{Ran}(B) = \overline{\text{Ran}(B)}$ i.e. the range of B is closed.

Proof. $\ker(B) = \{0\}$ follows from $\|x\| \leq c^{-1}\|Bx\|$ since $Bx = 0$ implies $x = 0$. Let $y \in \overline{\text{Ran}(B)}$ i.e. there exists $y_n \in Bx_n$ be Cauchy such that $y_n \rightarrow y$. But $\|x_n - x_m\| \leq c^{-1}\|Bx_n - Bx_m\| = c^{-1}\|y_n - y_m\| \rightarrow 0$. This implies $(x_n)_{n \in \mathbb{N}}$ is Cauchy. Since X is Banach, $x_n \rightarrow x \in X$.

$$Bx = B \lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} Bx_n = \lim_{n \rightarrow \infty} y_n = y$$

By definition of $\text{Ran}(B)$ we have $y \in \text{Ran}(B)$ i.e. $\text{Ran}(B) = \overline{\text{Ran}(B)}$. □

Corollary 1.60. Let B be the same as Lemma below and $X = Y$ then we have $\ker(B^n) = \{0\}$ and $\text{Ran}(B^n)$ is closed.

Proof. Since

$$\|B^n x\| = \|B(B^{n-1}X)\| \geq c\|B^{n-1}x\| \geq \dots \geq c^n\|x\|$$

□

Theorem 1.61. Let X be Banach and $T \in \text{Com}(X, Y)$ and $\lambda \neq 0$ is not an eigenvalue of T then $\lambda \in \rho(T)$.

Proof. We know that $\|(T - \lambda)x\| \geq c\|x\|$ for all $x \in X$ and $\ker(T - \lambda I)^n = \{0\}$ and $\text{Ran}(T - \lambda I)^n$ is closed. Let $X_n = \text{Ran}(T - \lambda I)^n \subset X$ and $X_0 = X$, observe that $X_{n+1} \subset X_n$ for all $n \in \mathbb{N}$. Then we know that $\dots \subset X_n \subset X_{n-1} \subset \dots \subset X_1 \subset X_0 := X$. We claim that the inclusions have to stop being proper at some $N \in \mathbb{N}$ i.e. there exists $N \in \mathbb{N}$ such that $X_{N+1} = X_N$.

Suppose the contradiction of the claim, i.e. all the inclusions are proper. Since X_1 is closed and $X_1 \neq X_0$ then there exists $x_0 \in X_0 \setminus X_1$ such that $\|x_0\| = 1$ and $\|x_0 - x_1\| \geq \frac{1}{2}$ for all $x_1 \in X_1$. Since X_2 is a closed subspace of X_1 so $X_1 \neq X_2$ then there exists $x_1 \in X_1 \setminus X_2$ such that $\|x_1\| = 1$ and $\|x_1 - x_2\| \geq \frac{1}{2}$ for all $x_2 \in X_2$. By induction, X_k is closed subspace of X_{k-1} and $X_k \neq X_{k-1}$ such that

$$\|x_k - x_{k'}\| \geq \frac{1}{2}, \quad \forall k < k'$$

Since $T \in \text{Com}(X, Y)$, $(Tx_n)_{n \in \mathbb{N}}$ has a Cauchy subsequence $(Tx_{n_j})_{j \in \mathbb{N}}$ then

$$\begin{aligned}Tx_{n_j} - Tx_{n_{j'}} &= Tx_{n_j} - \lambda x_{n_j} + \lambda x_{n_j} - Tx_{n_{j'}} + \lambda x_{n_{j'}} - \lambda x_{n_{j'}} \\ &= \lambda(x_{n_j} - x_{n_{j'}}) + (T - \lambda I)(x_{n_j}) - (T - \lambda I)(x_{n_{j'}}) \\ &= \lambda((x_{n_j} - x_{n_{j'}}) + \frac{1}{\lambda}((T - \lambda I)(x_{n_j}) - (T - \lambda I)(x_{n_{j'}})))\end{aligned}$$

Without loss of generality, assume $i < j$ then

$$\begin{aligned}\|Tx_{n_j} - Tx_{n_{j'}}\| &= |\lambda| \|(x_{n_j} - x_{n_{j'}}) + \frac{1}{\lambda}((T - \lambda I)(x_{n_j}) - (T - \lambda I)(x_{n_{j'}}))\| \\ &\geq \frac{|\lambda|}{2}\end{aligned}$$

a contradiction to T being compact. We know that $X_{N+1} = X_N = \text{Ran}((T - \lambda I)^N)$ and we want to show $X_N = X_{N-1}$, $X_N \subset X_{N-1}$ is trivial it is enough to show the other inclusion. Let $y \in X_{N-1}$, $y = (T - \lambda I)^{N-1}x$ for some $x \in X$ so $(T - \lambda I)y = (T - \lambda I)^N x = (T - \lambda I)^{N+1} \tilde{x}$. Then

$$(T - \lambda I)(y - (T - \lambda I)^N \tilde{x}) = 0$$

Since $\ker(T - \lambda I) = \{0\}$ we have

$$y - (T - \lambda I)^N \tilde{x} = 0 \implies y \in \text{Ran}(T - \lambda I)^N$$

Thus,

$$X_{N+1} = X_N = \dots = X_1 = X_0 = X$$

In particular, $\text{Ran}(T - \lambda I) = X$ and since λ is injective so $T - \lambda I$ is invertible. □

Definition 1.62. We say a system of vectors $\{e_\alpha\}_{\alpha \in J}$ is linearly independent if for any finite subset of $\{e_\alpha\}_{\alpha \in J}$ say $\{e_{\alpha_j}\}_{j=1}^N$

$$\sum_{j=1}^N c_j e_{\alpha_j} = 0 \implies c_1 = c_2 = \dots = c_N = 0$$

Lemma 1.63. If $\{e_\alpha\}_{\alpha \in J}$ is a set of eigenvectors corresponding to different eigenvalues, then $\{e_\alpha\}_{\alpha \in J}$ is linearly independent.

Proof. Suppose that N is the smallest number such that $(c_i)_{i=1}^N$ with $c_N \neq 0$ such that

$$\sum_{i=1}^N c_i e_{\alpha_i} = 0$$

Since $c_N \neq 0$,

$$e_{\alpha_N} = -\frac{1}{c_N} \sum_{i=1}^{N-1} c_i e_{\alpha_i}$$

And e_{α_i} are eigenvectors i.e. $Ae_{\alpha_N} = \lambda_N e_{\alpha_N}$. Apply A to both sides of the equation yields

$$\lambda_N e_{\alpha_N} = -\frac{1}{c_N} \sum_{i=1}^{N-1} c_i \lambda_i e_{\alpha_i}$$

$$\begin{aligned} \sum_{i=1}^{N-1} c_i \lambda_N e_{\alpha_i} + \lambda_N c_N e_{\alpha_N} &= 0 \\ \implies \sum_{i=1}^{N-1} c_i (\lambda_N - \lambda_i) e_{\alpha_i} &= 0 \end{aligned}$$

and we are done by contradiction. $\square$

Theorem 1.64. Let X be a Banach space and $T \in \text{Com}(X)$ then $\sigma(T)$ is at most countable and 0 is the only possible accumulating point. For all $0 \notin \lambda \in \sigma(T)$ is an eigenvalue with finite multiplicity (i.e. finite eigenspace).

Proof. It suffices to prove $\forall \delta > 0$,

$$|\sigma(T) \cap (B_\delta(0))^c| < \infty$$

Suppose the contradiction is true then there exists $\delta > 0$ such that $|\sigma(T) \cap (B_\delta(0))^c| < \infty$. Therefore, there exists (λ_n, x_n) such that

$$Tx_n = \lambda_n x_n \quad \text{and} \quad \lambda_n \in (B_\delta(0))^c$$

Consider $X_1 = \text{Span}(x_1)$ and $X_k = \text{Span}(x_k) \oplus X_{k-1}$ and we know that $X_k \neq X_{k+1}$. We know X_k is closed since it's finite dimensional. There exists $y_k \in X_k \setminus X_{k-1}$ such that $\|y_k\| = 1$ and $\|y_k - y_{k-1}\| \geq \frac{1}{2}$ for all $y_{k-1} \in X_{k-1}$ and $\|y_k - y_{k'}\| \geq \frac{1}{2}$ for $k > k'$.

Since T is compact $(Ty_k)_{k \in \mathbb{N}}$ has a Cauchy subsequence $(Ty_{k_m})_{m \in \mathbb{N}}$ then

$$\begin{aligned} \|Ty_{k_m} - Ty_{k_{m'}}\| &= \|\lambda y_{k_m} - \lambda y_{k_{m'}}\| \\ &\geq \frac{|\lambda_m|}{2} > \frac{\delta}{2} \end{aligned}$$

and we are done. $\square$

Definition 1.65. Let $(Y, \|\cdot\|_Y)$ be a normed space and X be a linear subspace of Y . Suppose X has another norm $\|\cdot\|_X$, we say X is continuously embedded into Y if $I : (X, \|\cdot\|_X) \rightarrow (Y, \|\cdot\|_Y)$ is continuous i.e. $\|Ix\|_Y \leq C\|x\|_X$ for $x \in X$.

If $I : (X, \|\cdot\|_X) \rightarrow (Y, \|\cdot\|_Y)$ is compact, then we say X is compactly embedded into Y , denoted by $X \subset\subset Y$.

Example. $C^1([0, 1])$ is continuously embedded into $C([0, 1])$ since

$$\|x\|_{C([0,1])} \leq \|x\|_{C^1([0,1])}$$

In particular, this embedding is in fact compact.

Proof. Let $(f_n)_{n \in \mathbb{N}}$ be a bounded sequence in $C^1([0, 1])$. We want to show that there exists $\|f_{n_j} - f_{n_{j'}}\|_{C([0,1])} \rightarrow 0$. Observe $\|f_n\|_{C([0,1])} = \sup_{x \in [0,1]} |f_n(x)| \leq \|f_n\|_{C^1([0,1])}$ thus $(f_n)_{n \in \mathbb{N}}$ is uniformly bounded. Then

$$\|f_n(x) - f_n(y)\|_{C([0,1])} = \sup |f_n(x) - f_n(y)| \leq \sup |f'_n(z)|(x - y)$$

so $(f_n)_{n \in \mathbb{N}}$ is uniformly equicontinuous. By Arzela-Ascoli theorem, there exists $(f_{n_j})_{j \in \mathbb{N}}$ is uniformly convergent i.e. for all $\delta > 0$, there exists $N \in \mathbb{N}$ such that

$$\|f_{n_j} - f_{n_{j'}}\|_{C([0,1])} < \delta, \quad \forall j, j' > N$$

$\square$

2 Linear Operators on Hilbert Space

Definition 2.1. An inner product space $(H, \langle \cdot, \cdot \rangle)$ satisfy:

- (i) $\langle x, y \rangle = \overline{\langle y, x \rangle}, \quad \forall x, y \in H$
- (ii) $\langle \alpha_1 x_1 + \alpha_2 x_2, y \rangle = \alpha_1 \langle x_1, y \rangle + \alpha_2 \langle x_2, y \rangle \quad \forall x, y \in H, \alpha_1, \alpha_2 \in \mathbb{F}.$
- (iii) $\langle x, x \rangle \geq 0$ and $\langle x, x \rangle = 0$ if and only if $x = 0$.

Examples.

- (1) $\mathbb{C}^N$ with $\langle x, y \rangle = \sum_{i=1}^N \omega_i x_i \overline{y_i}, \omega_i > 0.$
- (2) ℓ^2 with $\langle x, y \rangle = \sum_{i=1}^{\infty} x_i \overline{y_i}.$
- (3) $C([0, 1])$ with $\langle f, g \rangle = \int_0^1 f(t) \overline{g(t)} dt.$

Theorem 2.2 (Cauchy Schwarz).

$$|\langle x, y \rangle| \leq \|x\| \|y\|$$

Definition 2.3. An inner product space H with inner product $\langle \cdot, \cdot \rangle$ is called a Hilbert space if H is complete with respect to the induced norm $\| \cdot \|$.

Definition 2.4. A set of vectors $\{e_\alpha\}_{\alpha \in J}$ of an inner product space is called orthogonal if $\langle e_\alpha, e_\beta \rangle = 0$ for $\alpha \neq \beta$.

Theorem 2.5 (Pythagoras' theorem). If $\{e_\alpha\}_{\alpha \in J}$ is orthogonal then

$$\left\| \sum_{i=1}^N e_i \right\|^2 = \sum_{i=1}^N \|e_i\|^2$$

In general $\|x + y\|^2 = \|x\|^2 + 2 \operatorname{Re} \langle x, y \rangle + \|y\|^2$ and $\|x - y\|^2 = \|x\|^2 - 2 \operatorname{Re} \langle x, y \rangle + \|y\|^2$. Thus,

$$4 \operatorname{Re} \langle x, y \rangle = \|x + y\|^2 - \|x - y\|^2$$

If $\mathbb{F} = \mathbb{R}$ then this is called the polarization formula.

Theorem 2.6 (Jordan-Von Neumann). A inner product is induced by an inner product space if it satisfies parallelogram law:

$$\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2.$$

Example. ℓ^p is an inner product space if and only if $p = 2$.

Theorem 2.7. Let L be a closed subspace of a Hilbert space H for all $x \in H$ there exists unique $y \in L$ such that

$$\|x - y\| = \operatorname{dist}(x, L)$$

and $\langle x - y, y' \rangle = 0, \forall y' \in L$.

Proof. If $x \in L$, we can pick $y = x$ then $\operatorname{dist}(x, L) = 0$. Now suppose $x \notin L$, let y_n be a sequence in L minimizing $\operatorname{dist}(x, y_n) \rightarrow \operatorname{dist}(x, L)$ i.e.

$$\lim_{n \rightarrow \infty} \|x - y_n\| = \operatorname{dist}(x, L)$$

Since L is closed, $\frac{y_n + y_m}{2} \in L$ then

$$\operatorname{dist}(x, L) \leq \left\| x - \frac{y_n + y_m}{2} \right\| \implies 2\operatorname{dist}(x, L) \leq \|2x - (y_n + y_m)\|$$

$$\begin{aligned} \|y_n - y_m\| &= \|(y_n - x) + (x - y_m)\| \\ &= 2\|y_n - x\|^2 + 2\|x - y_m\|^2 - \|2x - (y_n + y_m)\|^2 \\ &\leq 2\|y_n - x\|^2 + 2\|x - y_m\|^2 - 4\operatorname{dist}(x, L) \rightarrow 0 \end{aligned}$$

] as $n, m \rightarrow \infty$. Thus, $(y_n)_{n \in \mathbb{N}}$ is Cauchy and let $y = \lim_{n \rightarrow \infty} y_n$ then we must have $\operatorname{dist}(x, L) = \|x - y\|$. Suppose there exists $y_0 \in L$ such that $\langle x - y, y_0 \rangle \neq 0$. Without loss of generality we assume $\operatorname{Re} \langle x - y, y_0 \rangle > 0$ and $\|y_0\| = 1$. Want to show $y + \alpha y_0$ is a minimizer of some small $\alpha \in \mathbb{R}$.

$$\|x - (y + \alpha y_0)\|^2 = \|(x - y) - \alpha y_0\|^2 = \|x - y\|^2 - 2\alpha \operatorname{Re} \langle x - y, y_0 \rangle + \alpha^2 < \|x - y\|^2$$

by choosing α small enough. This shows that $\langle x - y, u' \rangle = 0$ for all $u' \in L$. Suppose y_1 is another minimizer then $\langle x - y_1, z \rangle = 0$ and $\langle x - y, z \rangle = 0$ for all $z \in L$ which implies

$$\langle y - y_1, z \rangle = 0$$

for $z \in L$. Let $z = y - y_1 \in L$ then we have $y = y_1$. □

Corollary 2.8. Let X_0 be a closed subspace of a Hilbert space H and $X_0 \neq H$. Then there exists $x \in H \setminus X_0$, with $\|x\| \leq 1$ such that $\|x - x_0\| \geq 1$.

Definition 2.9. Let

$$\text{lin}(A) = \left\{ \sum_{i=1}^N c_i x_i, \quad c_i \in \mathbb{F}, \lambda_i \in A, N \in \mathbb{N} \right\}$$

and

$$\text{Span}(A) = \overline{\text{lin}(A)}$$

Example. $e_n = (0, \dots, 1, 0, \dots) \in \ell^\infty$ with 1 in the n -th position. Let $A : (e_n)_{n \in \mathbb{N}}$, then $\text{lin}(A) = c_{00}$ and $\text{Span}(A) = \overline{c_{00}}^{\ell^\infty} = c_0$.

Theorem 2.10. If H is finite dimensional, $A \subset H$ then $\text{lin}(A) = \text{Span}(A)$.

Definition 2.11. Let M be a subset of a Hilbert space H , the annihilator of M is defined as

$$M^\perp = \{y \in H \mid \langle x, y \rangle = 0 \quad \forall x \in M\}$$

Proposition 2.12.

- (1) $M^\perp = \overline{M^\perp}$.
- (2) $M_1 \subset M_2 \implies M_2^\perp \implies M_1^\perp$.
- (3) $M \subset M^{\perp\perp}$.
- (4) $M^\perp \subset \text{Span}(M)^\perp$.
- (5) $M^{\perp\perp} = \text{Span}(M)$.

Theorem 2.13. Let $\{e_\alpha\}_{\alpha \in \mathbb{N}}$ be an orthonormal set of a Hilbert space H then the following statements are equivalent:

- (1) $x = \sum_{i=1}^\infty \langle x, e_i \rangle e_i$ for all $x \in H$.
- (2) $\|x\|^2 = \sum_{i=1}^\infty |\langle x, e_i \rangle|^2$
- (3) $\langle x, e_i \rangle = 0$ for $i \in \mathbb{N}$ implies $x = 0$.
- (4) $\text{lin}(\{e_i\}_{i \in \mathbb{N}})$ is dense in H .

Definition 2.14. A normed space is called separable if it contains a countable dense set.

Theorem 2.15. A Hilbert space is separable if and only if it contains a countable complete orthonormal set.

Example. $L^2(0, 1)$ with $\langle f, g \rangle = \int_0^1 f \bar{g} dt$ and $e_n(t) = e^{2\pi i n t}$ with $n = 0, 1, \dots$ is complete orthonormal set.

Theorem 2.16. Let $U : H_1 \rightarrow H_2$ be an isomorphism i.e. $(\|Ux\|_{H_2} = \|x\|_{H_1})$ by polarization formula we have

$$\begin{aligned} 4\langle Ux, Uy \rangle_{H_2} &= \|U(x+y)\|_{H_2}^2 - \|U(x-y)\|_{H_2}^2 \\ &= \|x+y\|_{H_1}^2 - \|x-y\|_{H_1}^2 \\ &= 4\langle x, y \rangle \end{aligned}$$

Theorem 2.17 (Riesz Representation Theorem). f is a bounded linear functional on a Hilbert space H if and only there exists unique $y \in H$ such that $f(x) = \langle x, y \rangle$ for all $x \in H$.

Corollary 2.18. $f \in H^* = \mathcal{B}(H, \mathbb{F}) \cong H$.

Definition 2.19. Adjoint of $B \in \mathcal{B}(H)$ is defined by B^* satisfying

$$\langle Bx, y \rangle = \langle x, B^*y \rangle \quad \forall x, y \in H$$

Proposition 2.20.

- (1) $(\alpha A + \beta B)^* = \bar{\alpha} A^* + \bar{\beta} B^*$.

- (2) $A^{**} = A$.
- (3) $(AB)^* = B^*A^*$.
- (4) $\|B^*\| = \|B\|$.
- (5) $\|B^*B\| = \|BB^*\| = \|B\|^2$.
- (6) B is invertible iff B^* is invertible.

Theorem 2.21. Let H_1, H_2 be Hilbert spaces and $B \in \mathcal{B}(H_1, H_2)$ then $\ker(B^*) = \text{Ran}(B)^\perp$ and $\ker(B) = \text{Ran}(B^*)^\perp$.

Proof. $x \in \ker(B^*)$ if and only if $B^*x = 0$ if and only if $\langle B^*y, x \rangle_{H_1} = 0$ for all $y \in H_1$ if and only if $\langle x, By \rangle = 0$ for all $y \in H_1$ if and only if $x \in \text{Ran}(B)^\perp$. Other equality is similar. $\square$

Corollary 2.22. $\overline{\text{Ran}(B)} = \ker(B^*)^\perp$, $\overline{\text{Ran}(B^*)} = \ker(B)^\perp$.

Definition 2.23. Let $B \in \mathcal{B}(X, Y)$ then $B^* \in \mathcal{B}(Y^*, X^*)$ then B^*B makes sense when $Y^* = Y$ and BB^* makes sense when $X^* = X$. And

$$B^*B \in \mathcal{B}(X, X^*), \quad BB^* \in \mathcal{B}(Y^*, Y)$$

When $X^* = Y$ and $Y^* = X$ then $B^*B - BB^*$ makes sense. In particular when $X = Y = H$ with H be a Hilbert space and $B \in \mathcal{B}(H)$. We say

- (1) B is normal if $B^*B = BB^*$.
- (2) B is self-adjoint if $B = B^*$.
- (3) $U \in \mathcal{B}(H_1, H_2)$ we say U is unitary if $U^*U = I_{H_1}$ and $UU^* = I_{H_2}$ i.e. $U^* = U^{-1}$.

Remark. Self-adjoint operators are normal. If $H_1 = H_2$ then unitary operators are also normal.

Theorem 2.24. Let $B \in \mathcal{B}(H)$, H is a Hilbert space. Then B is normal if and only if

$$\|Bx\| = \|B^*x\|, \quad \forall x \in H$$

Proof. ($\implies$) : If B is normal, then

$$\|Bx\|^2 := \langle Bx, Bx \rangle = \langle B^*Bx, x \rangle = \langle BB^*x, x \rangle = \langle B^*x, B^*x \rangle$$

($\impliedby$) : We want to show $(BB^* - B^*B)x = 0$ for all $x \in H$. Note that

$$\begin{aligned} 4\langle Bx, By \rangle &= \|B(x+y)\|^2 + \|B(x-y)\|^2 + i(\dots) \\ &= \|B^*(x+y)\|^2 + \|B^*(x-y)\|^2 + i(\dots) \\ &= 4\langle B^*x, B^*y \rangle \end{aligned}$$

$$\implies \langle B^*Bx, y \rangle - \langle BB^*x, y \rangle = 0 \quad \forall x, y \in H$$

Let $y = (B^*B - BB^*)x$ then we conclude that $(B^*B - BB^*)x = 0$. $\square$

Theorem 2.25. Let $B \in \mathcal{B}(H)$ be normal.

- (1) $\text{Ran}(B^*)^\perp = \ker(B) = \ker(B^*) = \text{Ran}(B)^\perp$.
- (2) $Bx = \alpha x \iff B^*x = \bar{\alpha}x$.
- (3) Eigenvectors for different eigenvalues are orthogonal.

Proof.

- (1) By previous Theorem, B is normal implies

$$\begin{aligned} \|Bx\| &= \|B^*x\| \quad \forall x \\ \implies \ker(B) &= \ker(B^*) \end{aligned}$$

- (2) Note that

$$\begin{aligned} \|(B - \alpha)x\|^2 &= \langle (B - \alpha)x, (B - \alpha)x \rangle \\ &= \langle Bx, Bx \rangle - \langle Bx, \alpha x \rangle - \langle \alpha x, Bx \rangle + \langle \alpha x, \alpha x \rangle \\ &= \|B^*x\|^2 - \langle \bar{\alpha}x, B^*x \rangle - \langle B^*x, \bar{\alpha}x \rangle + \|\bar{\alpha}x\|^2 \\ &= \|(B^* - \bar{\alpha})x\|^2 \end{aligned}$$

- (3) Let $\lambda_i \neq \lambda_j$ and $Bx_i = \lambda_i x_i$ and $Bx_j = \lambda_j x_j$. Then

$$\begin{aligned} (\lambda_i - \lambda_j)\langle x_i, x_j \rangle &= \langle Bx_i, x_j \rangle - \langle x_i, \bar{\lambda}_j x_j \rangle \\ &= \langle Bx_i, x_j \rangle - \langle x_i, B^*x_j \rangle \\ &= \langle (B - B^*)x_i, x_j \rangle = 0 \end{aligned}$$

$\square$

Theorem 2.26. $U \in \mathcal{B}(H_1, H_2)$, then the following statements are equivalent

- (1) U is unitary.
- (2) $\text{Ran}(U) = H_2$ and $\langle Ux, Uy \rangle = \langle x, y \rangle$ for all $x, y \in H_1$.
- (3) $\text{Ran}(U) = H_2$ and $\|Ux\| = \|x\|$ for all $x \in H_1$.

Proof. (1 $\implies$ 2) : This means $UU^* = I_{H_2}$ which implies $\text{Ran}(U) = H_2$ and

$$\langle Ux, Uy \rangle = \langle U^*Ux, y \rangle = \langle I_{H_1}x, y \rangle = \langle x, y \rangle$$

(2 $\implies$ 1) : $\langle U^*Ux, y \rangle = \langle x, y \rangle$ for all $x, y \in H_1$.

$$\langle (U^*U - I_{H_1})x, y \rangle = 0 \implies (U^*U - I_{H_1})x = 0$$

Similarly we can show that $UU^* = I_{H_2}$. □

Theorem 2.27.

- (1) B is self-adjoint and $\lambda \in \mathbb{R}$ implies λB is self-adjoint.
- (2) B_1, B_2 are self-adjoint then $B_1 + B_2$ are self-adjoint.
- (3) If B_1, B_2 are self-adjoint then B_1B_2 is self-adjoint if $[B_1, B_2] = 0$.
- (4) $(B_n)_{n \in \mathbb{N}}$ are sequence of self-adjoint operators and $\|B - B_n\| \rightarrow 0$ as $n \rightarrow \infty$ then B is self-adjoint.

Proof. We shall only prove statement(4) which is non-trivial. For all $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$\|B - B_n\| < \frac{\varepsilon}{2} \implies \|B^* - B_n^*\| < \frac{\varepsilon}{2}$$

then

$$\begin{aligned} 0 \leq \|B - B^*\| &= \|(B - B_n) - (B^* - B_n^*)\| \\ &\leq \|B - B_n\| + \|B^* - B_n^*\| \leq \varepsilon \end{aligned}$$

thus $B = B^*$. □

Theorem 2.28. $B \in \mathcal{B}(X)$ is self-adjoint if and only if

$$\langle Bx, x \rangle = \overline{\langle Bx, x \rangle} = \langle x, Bx \rangle \quad \forall x \in H$$

Proof. ($\implies$) : Clear.

($\impliedby$) : Note that

$$\langle B(x+y), x+y \rangle = \langle Bx, x \rangle + \langle Bx, y \rangle + \langle By, x \rangle + \langle By, y \rangle$$

and

$$\langle B(ix+y), ix+y \rangle = \langle Bx, x \rangle + i\langle Bx, y \rangle - i\langle By, x \rangle + \langle By, y \rangle$$

Then

$$\langle B(x+y), x+y \rangle = \langle Bx, x \rangle + \langle y, Bx \rangle + \langle x, By \rangle + \langle By, y \rangle$$

Therefore we have

$$\langle Bx, y \rangle - \langle x, By \rangle + \langle By, x \rangle - \langle y, Bx \rangle = 0$$

Similarly we have

$$i(\langle Bx, y \rangle - \langle By, x \rangle) - i(\langle x, By \rangle - \langle y, Bx \rangle) = 0$$

Rerranging all the terms gives $\langle Bx, y \rangle = \langle x, By \rangle$ for all $x, y \in H$. □

Theorem 2.29. $|\langle Bx, x \rangle| \geq c\|x\|^2$ implies B is invertible.

Definition 2.30. Let $B \in \mathcal{B}(H)$, the numerical range of B is given by

$$\text{Num}(B) := \{\langle Bx, x \rangle \mid \|x\| = 1, \quad \forall x \in H\}$$

Theorem 2.31. Let H be a Hilbert space and $B \in \mathcal{B}(H)$. Then

$$\sigma(B) \subset \overline{\text{Num}(B)}$$

Theorem 2.32. Let H be a Hilbert space, $B \in \mathcal{B}(H)$ be self-adjoint. Then

$$\|B\| = \sup_{\|x\|=1} |\langle Bx, x \rangle|$$

Proof. Since $|\langle Bx, x \rangle| \leq c$ and replace $x = \frac{y}{\|y\|}$ then we have

$$|\langle Bx, x \rangle| \leq c\|x\|^2$$

Let $c = \sup_{\|x\|=1} |\langle Bx, x \rangle|$ and $c \leq \|B\|$. Note that

$$\begin{aligned}\langle B(x+y), (x+y) \rangle &= \langle Bx, x \rangle + \langle Bx, y \rangle + \langle By, x \rangle + \langle By, y \rangle \\ &= \langle Bx, x \rangle + 2\operatorname{Re}\langle Bx, y \rangle + \langle By, y \rangle\end{aligned}$$

And

$$\begin{aligned}\langle B(x-y), (x-y) \rangle &= \langle Bx, x \rangle - 2\operatorname{Re}\langle Bx, y \rangle + \langle By, y \rangle \\ \implies 4\operatorname{Re}\langle Bx, y \rangle &= \langle B(x+y), (x+y) \rangle - \langle B(x-y), (x-y) \rangle \\ \langle B(x+y), (x+y) \rangle - \langle B(x-y), (x-y) \rangle &\leq c(\|x+y\|^2 + \|x-y\|^2) \\ &\leq 2c(\|x\|^2 + \|y\|^2)\end{aligned}$$

Thus, $4\operatorname{Re}\langle Bx, y \rangle \leq 4c$ for $x, y \in H$ and $\|x\| = \|y\| = 1$. Let $y = \frac{Bx}{\|Bx\|}$ which implies

$$\begin{aligned}\frac{\langle Bx, Bx \rangle}{\|Bx\|} &\leq c \text{ for } x \in H, \|x\| = 1 \\ \implies \|B\| &\leq c\end{aligned}$$

□

Theorem 2.33. Let H be a Hilbert space and $B \in \mathcal{B}(H)$ be self-adjoint let

$$m := \inf \langle Bx, x \rangle, \quad M := \sup \langle Bx, x \rangle$$

Then

- (1) $\sigma(B) \subset [m, M] \subset \mathbb{R}$.
- (2) $\{m, M\} \subset \sigma(B)$.

Proof.

- (1) We have seen before that $\sigma(B) \subset \overline{\operatorname{Num}(B)} = [m, M] \subset \mathbb{R}$.
- (2) Let $\alpha = \frac{M+m}{2}$ and $\beta = \frac{M-m}{2}$ and $A = B - \alpha$ then A is self-adjoint and $\beta \geq 0$.

$$\inf_{\|x\|=1} \langle Ax, x \rangle = \inf_{\|x\|=1} \langle Bx, x \rangle - \alpha = m - \frac{M+m}{2} = -\beta$$

Similarly we have

$$\sup_{\|x\|=1} \langle Ax, x \rangle = \sup_{\|x\|=1} \langle Bx, x \rangle - \alpha = M - \frac{M+m}{2} = \beta$$

And $\beta = \|A\|$. There exists sequence $(x_n)_{n \in \mathbb{N}}$ with $\|x_n\| = 1$ such that $\lim_{n \rightarrow \infty} \langle Ax_n, x_n \rangle = \beta$.

$$\begin{aligned}\langle (A - \beta I)x_n, (A - \beta I)x_n \rangle &= \langle Ax_n, Ax_n \rangle - \beta \langle x_n, Ax_n \rangle - \beta \langle Ax_n, x_n \rangle + \beta^2 \\ &= \|Ax_n\|^2 - 2\beta \langle Ax_n, x_n \rangle + \beta^2 \\ &\leq \|A\|^2 - 2\beta \langle Ax_n, x_n \rangle + \beta^2 \\ &= 2\beta^2 - 2\beta \langle Ax_n, x_n \rangle \rightarrow 0\end{aligned}$$

i.e. $\|(A - \beta I)x_n\| \rightarrow 0$ as $n \rightarrow \infty$. We want to show $\beta \in \sigma(A)$. Suppose $A - \beta I$ is invertible then $R(A - \beta I) = I$ and $\|R(A - \beta I)x_n\| \leq \|R\|\|(A - \beta I)x_n\|$ but we see that $\|(A - \beta I)x_n\| \rightarrow 0$ contradicts to $\|x_n\| = 1$. Thus, $A - \beta I$ is not invertible which shows $\beta \in \sigma(A)$ i.e. $\beta \in \sigma(B - \alpha)$. By Spectral Mapping Theorem we have $M = \alpha + \beta \in \sigma(B)$. Similarly, we can show that $m \in \sigma(B)$.

□

Corollary 2.34. $r(B) = \|B\|$.

Theorem 2.35. If $U \in \mathcal{B}(H_1, H_2)$ is unitary then

$$\sigma(U) \subset \{\lambda \in \mathbb{C} : |\lambda| = 1\}$$

Proof. By definition, the unitary operator

$$\|Ux\| = \|x\| \quad \forall x \in H_1 \implies \|U\| = 1$$

$\sigma(U) \subset \{\lambda \in \mathbb{C} : |\lambda| \leq \|U\| = 1\}$, let $|\lambda| < 1$ then $U - \lambda I = U(I - \lambda U^{-1})$. But $\|\lambda U^{-1}\| \leq |\lambda|\|U^{-1}\| = |\lambda| < 1$. $I - \lambda U^{-1}$ is invertible which implies $U - \lambda I$ is invertible. □

Theorem 2.36. Let H be a Hilbert space and $B \in \mathcal{B}(H)$ be normal then $r(B) = \|B\|$.

Proof. Note that B is normal if and only if

$$\begin{aligned}\|Bx\| &= \|B^*x\|, \quad \forall x \in H \\ \implies \|BBx\| &= \|B^*Bx\| \text{ for } x \in H \implies \|B^2\| = \|B^*B\|\end{aligned}$$

and we have shown that $\|B^*B\| = \|B\|^2$ thus

$$\|B^2\| = \|B\|^2$$

$$\implies \|B^2 \cdot B^2\| = \|B^2\|^2 = \|B\|^4$$

i.e. $\|B^{2^k}\| = \|B\|^{2^k}$ for all k . Recall

$$r(B) = \lim_{n \rightarrow \infty} \|B^n\|^{1/n} = \lim_{k \rightarrow \infty} \|B^{2^k}\|^{1/2^k} = \|B\|$$

□

Theorem 2.37. Let H be a Hilbert space and $B \in \mathcal{B}(H)$ be normal then

$$\|B\| = \sup_{\|x\|=1} |\langle Bx, x \rangle|$$

Proof. By definition,

$$\begin{aligned} r(B) &= \sup\{|\lambda| : \lambda \in \sigma(B)\} \leq \sup\{|\lambda| : \lambda \in \overline{\text{Num}(B)}\} \\ &= \sup\{|\lambda| : \lambda \in \text{Num}(B)\} = \sup\{|\langle Bx, x \rangle| : \|x\| = 1, x \in H\} \\ &\leq \sup\{\|Bx\| : \|x\| = 1, x \in H\} = \|B\| \end{aligned}$$

By previous Theorem, we see that $\|B\| \leq r(B) \leq \|B\|$. Thus,

$$\|B\| = \sup_{\|x\|=1} |\langle Bx, x \rangle|$$

□

Theorem 2.38. Let X be a normed space, H be a Hilbert space and $T \in \text{Com}(X, H)$. Then there exists $(T_n)_{n \in \mathbb{N}}$, T_n being finite rank operators such that $\|T - T_n\| \rightarrow 0$ as $n \rightarrow \infty$.

Proof. Since T is compact, we know that $T(S_X)$ is relatively compact therefore there exists $y_1, \dots, y_{N(n)}$ such that $T(S_X) \subset \bigcup_{i=1}^{N(n)} B_{1/n}(y_i)$. Let $L = \text{lin}\{y_i\}_{i=1}^{N(n)}$ and define P_n to be orthogonal projection onto L . $y = P_n y + z$ with $P_n y \in L$ and $z \in L^\perp$ thus $Tx = P_n T x + z$.

$$\begin{aligned} \|Tx - P_n T x\| &= \text{dist}(Tx, L) < \frac{1}{n} \text{ for all } x \in S_X \\ \implies \|T - P_n T\| &< \frac{1}{n} \end{aligned}$$

And $T_n = P_n T$.

□

Let H_1, H_2 be separable Hilbert space. We can find $\{e_i\}_{i=1}^N$ and $\{f_j\}_{j=1}^M$ with $N, M \in \mathbb{N} \cup \{\infty\}$. Let $T \in \mathcal{B}(H_1, H_2)$ and

$$\begin{aligned} \sum_{i=1}^N \|Te_i\|^2 &= \sum_{i=1}^N \langle Te_i, Te_i \rangle \\ &= \sum_{i,j}^{M,N} \langle Te_i, f_j \rangle \langle f_j, Te_i \rangle \\ &= \sum_{i,j}^{M,N} \langle e_i, T^* f_j \rangle \langle T^* f_j, e_i \rangle \\ &= \sum_{j=1}^M \|T^* f_j\|^2 \end{aligned}$$

In particular, $\sum_{i=1}^\infty \|Te_i\|^2 = \sum_{j=1}^\infty \|T^* f_j\|^2$. In fact, it's independent of basis we choose.

Definition 2.39. $T \in \mathcal{B}(H_1, H_2)$ is called Hilbert-Schmidt if

$$\|T\|_{HS} := \sqrt{\sum_{i=1}^\infty \|Te_i\|^2} < \infty$$

Proposition 2.40. Hilbert-Schmidt operators is a subspace of $\mathcal{B}(H_1, H_2)$ i.e.

$$\alpha_1 T_1 + \alpha_2 T_2 \in H.S.$$

if $T_1, T_2 \in H.S.$ and $\alpha_1, \alpha_2 \in \mathbb{F}$.

Proof. Note that

$$\|\alpha T\|_{HS} = \sqrt{\sum_{i=1}^\infty \|\alpha Te_i\|^2} = |\alpha| \|T\|_{HS}$$

and

$$\begin{aligned}\|T_1 + T_2\|_{HS}^2 &= \sum_{i=1}^{\infty} \|(T_1 + T_2)e_i\|^2 \leq 2 \sum_{i=1}^{\infty} \|T_1 e_i\|^2 + \|T_2 e_i\|^2 \leq 2\|T_1\|_{HS}^2 + 2\|T_2\|_{HS}^2 \\ &\implies \|T_1 + T_2\|_{HS} \leq c(\|T_1\|_{HS} + \|T_2\|_{HS})\end{aligned}$$

i.e. $\|T_1 + T_2\|_{HS} < \infty$. □

Theorem 2.41. If $T \in \mathcal{B}(H_1, H_2)$ is Hilbert-Schmidt then T is compact.

Proof. Define $P_N y = \sum_{i=1}^N \langle y, f_j \rangle f_j$ observe $P_N T$ is a finite rank operator. We want to show $\|T - P_N T\| \rightarrow 0$ as $N \rightarrow \infty$. Note

$$\begin{aligned}\|Ty - P_N Ty\|^2 &= \left\| \sum_{j=N+1}^{\infty} \langle Ty, f_j \rangle f_j \right\|^2 \\ &\leq \sum_{j=N+1}^{\infty} |\langle Ty, f_j \rangle|^2 \\ &= \sum_{j=N+1}^{\infty} |\langle y, T^* f_j \rangle|^2 \\ &\leq \sum_{j=N+1}^{\infty} \|y\|^2 \|T^* f_j\|^2 \\ &\leq \|y\|^2 \sum_{j=N+1}^{\infty} \|T^* f_j\|^2\end{aligned}$$

Since $\|T\|_{HS} = \sum_{j=1}^{\infty} \|T^* f_j\|^2 < \infty$. i.e. $\|T - P_N T\| \rightarrow 0$ as $N \rightarrow \infty$. T is compact by property of compact operators. □

Theorem 2.42. Let H be a Hilbert space then $B \in \mathcal{B}(H)$ be a compact self-adjoint operator. Then there exists a finite or countable orthonormal set $\{e_i\}_{i=1}^N$, $N \in \mathbb{N} \cup \{\infty\}$ such that for all $x \in H$,

$$x = \sum_{i=1}^N c_i e_i + y$$

and $y \in \ker(B)$. Then

$$Bx = \sum_{i=1}^N \lambda_i c_i e_i$$

and $\sigma(B) \setminus \{0\} = \{\lambda_i\}_{i=1}^N$, $N \in \mathbb{N} \cup \{\infty\}$. Moreover, $|\lambda_1| \geq |\lambda_2| \geq \dots$. If $N = \infty$, then $\lim_{n \rightarrow \infty} \lambda_n = 0$.

Proof. We know that $\sigma(B) \cap \{|\lambda| > \delta\}$ is finite. For $\lambda \in \sigma(B) \cap \{|\lambda| > \delta\}$ we can order them which gives λ_i . Then we can use Gram-Schmidt process to find the orthonormal basis for eigenspace for λ_i . This implies there exists $(e_j)_{j=1}^N$ that corresponds to $\lambda_i \in \sigma(B) \setminus \{0\}$. Let $L = \text{Span}\{e_i\}_{i=1}^N$. We want to show $H = \ker(B) \oplus L$ i.e. $L^\perp = \ker(B)$. Let $y \in \ker(B)$ then we have

$$\langle y, e_i \rangle = \frac{1}{\lambda_i} \langle y, \lambda_i e_i \rangle = \frac{1}{\lambda_i} \langle By, e_i \rangle = 0$$

i.e. $y \in L^\perp$. If $y \in L^\perp$ then

$$0 = \langle y, e_i \rangle = \frac{1}{\lambda_i} \langle By, e_i \rangle$$

$By \in L^\perp$. Thus, $B : L^\perp \rightarrow L^\perp$. $\tilde{B} = B|_{L^\perp} : L^\perp \rightarrow L^\perp$ is also compact which implies any eigenvector of $\tilde{B}$ is also an eigenvector of B . This implies $\tilde{B} = 0$ $\tilde{B}y = 0$ i.e. $y \in \ker(B)$. □